



Mg-coordinated self-assembly of MgO-doped ordered mesoporous carbons for selective recovery of phosphorus from aqueous solutions

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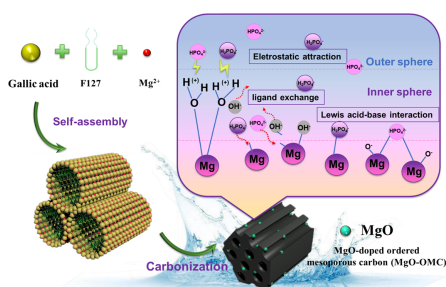
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HIGHLIGHTS

- One-pot Mg-coordinated self-assembly of MgO-doped ordered mesoporous carbons.
- Green process for synthesis of OMC-MgO from biomass-derived gallic acid.
- Mg acts as both crosslinker in OMC preparation and effective P sorption sites.
- OMC-MgO has high capacity for phosphate recovery from actual wastewater.
- OMC-MgO adsorbent is recyclable and reusable.

GRAPHICAL ABSTRACT



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ABSTRACT

MgO-doped ordered mesoporous carbon (OMC-MgO) was synthesized by evaporation-induced self-assembly (EISA) in one-pot with biomass-derived gallic acid instead of phenolics as carbon precursor, metal ion Mg^{2+} instead of formaldehyde as cross-linking agent and F127 as template. Mg had dual roles as cross-linking agent to coordinate gallic acid-F127 micelle composites, and as promoter of active sites. With the developed protocol, MgO could be uniformly dispersed throughout the carbon skeleton during calcination at 800 °C without agglomeration. As-prepared OMC-MgO-T800 had a specific surface area of 808 m²/g, a perfectly-ordered mesoporous structure centered at (6–8) nm, and exhibited good performance for phosphorus removal from aqueous solutions with a maximum sorption capacity of 107 mg/g. Electrostatic attraction, ligand exchange, and Lewis acid-base interactions are the main phosphate adsorption mechanisms of the developed MgO-doped OMC adsorbents. The as-prepared mesoporous carbon materials are highly selective for phosphorus adsorption from aqueous streams and are reusable and recyclable. This work proposes a green protocol to synthesize OMC-metal oxide composites that is widely applicable to wastewater treatment and technological areas in catalysis and energy storage.

1. Introduction

As one of the essential elements in biological growth and reproduction [1–3], excessive release of phosphorus (P) into aquatic

ecosystems promotes explosive growth of harmful algal blooms, thereby deteriorating water quality and become fatal to aquatic animals and eventually destroying ecological balances (*i.e.* eutrophication) [4–7]. Environmental protection agencies in countries around the

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